Angles in the Same Segment

Exam code: 0607

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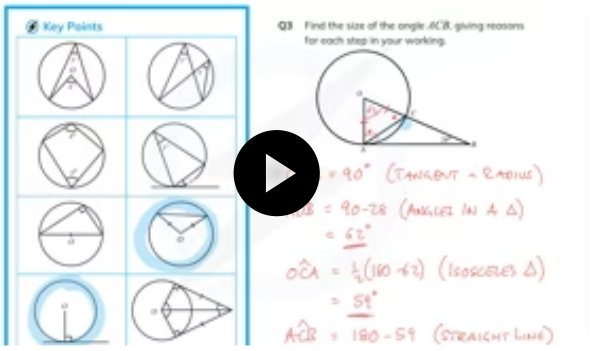

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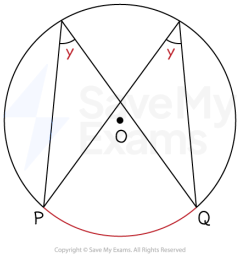
Help me understand

Circles & segments

Circle Theorem: Angles in the same segment are equal

- **Any two angles** on the **circumference** of a circle that are formed from the **same two points** on the circumference are **equal**
  - These two angles are in the **same segment** of the circle
    - To see this, add the chord *PQ* below to split the circle into two segments

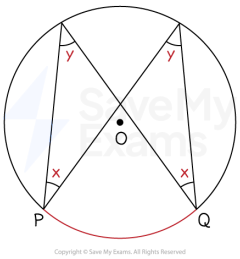


- To spot this circle theorem on a diagram
  - Find **two points** on the **circumference** that meet at a **third point**
  - See if there are **any other pairs of lines** from the **same two original points** that meet at a different point on

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **Angles in the same segment are equal**

- Look out for a **bowtie shape**
  - The theorem works **upside down**, in that the angles at *P* and *Q* are **also equal**



Examiner Tips and Tricks

An exam question diagram may have **multiple equal angles**.

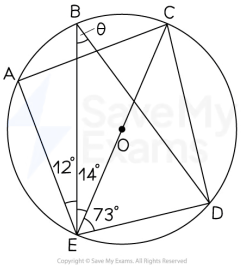
Look for as many as possible by seeing how many pairs of lines start from the same two points on the circumference.

Worked Example

The diagram below shows a circle with centre, *O*.
A, *B*, *C*, *D* and *E* are five points on the circumference on the circle.

Angle *AEB* = 12°.
Angle *BEC* = 14°.
Angle *CED* = 73°.
Angle *EBD* = θ°.

Find the value of θ.



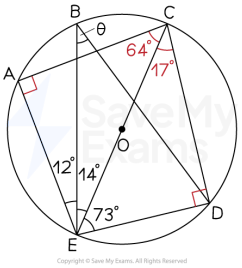
Answer:

CE is a diameter
This means that triangles *EAC* and *CED* are both triangles in a semicircle

Angle *EAC* = 90°
Angle *CED* = 90°

Angle $ECA = 64°$
Angle $ECD = 17°$
Angles in a triangle = 180°

Label these angles on the diagram



Angle θ is formed by two lines coming from either end of the chord ED
Angle ECD is also formed by two lines coming from either end of the chord ED

Angle θ = angle ECD = 17°
Angles in the same segment are equal

Angle θ = 17°

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